

TrES-3b

R-1270

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160430 · created 2026-07-31T13:49:59+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457508.8486 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.012
Transit depth (Rp/Rs)^2	3.82 +/- 0.48 [%]
Orbital Inclination (inc)	81.9 +/- 0.32
Airmass coefficient 1 (a1)	1.284 +/- 0.014
Airmass coefficient 2 (a2)	-0.2313 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	None
Optimal Aperture	3.43
Optimal Annulus	4.22
Transit Duration (day)	0.0591 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.012 vs 0.16309 (deviation 1.94σ)
Duration fit vs published	0.0591 d vs 0.0567 d (deviation 0.57σ)
Mid-time O–C	-1.4 min
Depth SNR	11.5
Residual scatter	1.09 %

Light curve

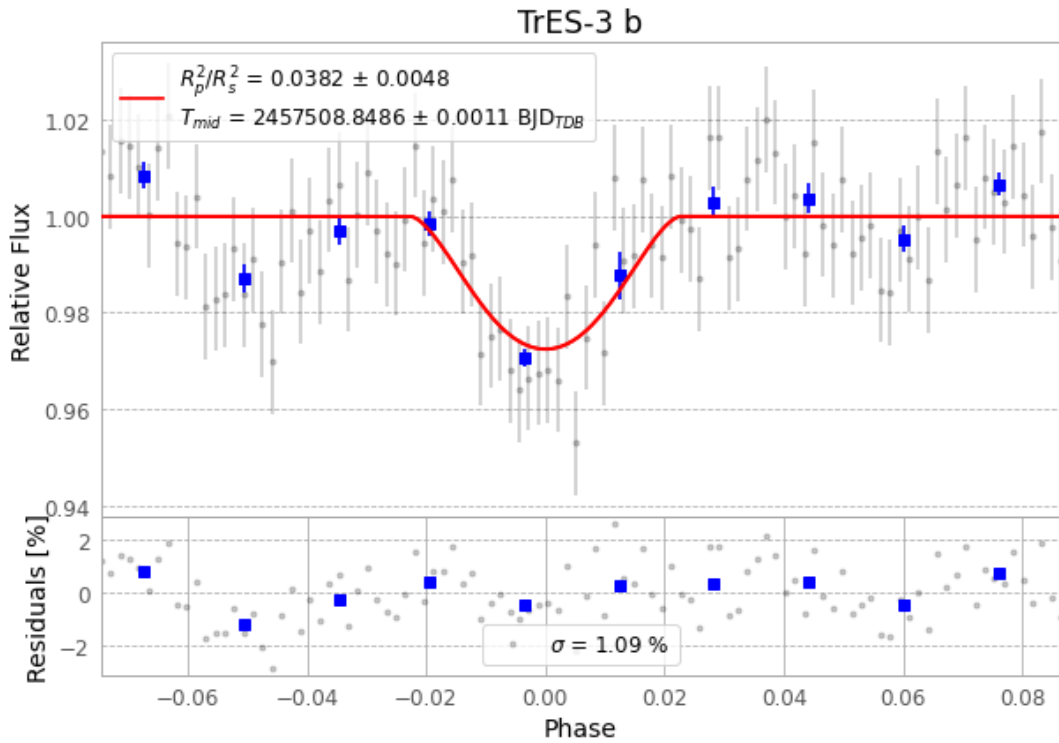


Figure 1 — FinalLightCurve_TrES-3 b_2016-04-29.png (SHA-256 46de6a66a9d9add5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [296, 166], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 290d7520ffd2390c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

102 FITS frames (67 MB), TRES-3160430055712.FITS → TRES-3160430110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160430.log (1102bbe95b9f...), FinalLightCurve_TrES-3 b_2016-04-29.png (46de6a66a9d9...), FinalLightCurve_TrES-3 b_2016-04-29.csv (382ea9381531...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 13:50Z.